MOBSF

ANDROID STATIC ANALYSIS REPORT



WaterSOS (1.0)

File Name:	app-debug.apk
Package Name:	com.example.watersos
Scan Date:	Oct. 21, 2026, 4:38 p.m.
App Security Score:	35/100 (HIGH RISK)
Grade:	C

FINDINGS SEVERITY

HIGH	MEDIUM	INFO	SECURE	HOTSPOT
5	6	0	1	1

FILE INFORMATION

File Name: app-debug.apk
Size: 5.17MB
MD5: 24c1fd8bcf53f1a15dd1199b9bbcfc85
SHA1: 99041be89349ac77e55bea3ede7ee3f3bd07a51c
SHA256: 13193e09ef8add7f3a6cd3752dcd2bccf7f363d77cc38b971b4d219a5b8ea82b

APP INFORMATION

App Name: WaterSOS
Package Name: com.example.watersos
Main Activity: com.example.watersos.Activities.PresentacionActivity
Target SDK: 30
Min SDK: 23
Max SDK:
Android Version Name: 1.0
Android Version Code: 1

APP COMPONENTS

Activities: 5
Services: 0
Receivers: 0
Providers: 1
Exported Activities: 1
Exported Services: 0
Exported Receivers: 0
Exported Providers: 0

CERTIFICATE INFORMATION

Binary is signed
v1 signature: True
v2 signature: True
v3 signature: False
v4 signature: False
v3.1 signature: False
v3.2 signature: False
X.509 Subject: CN=Android Debug, O=Android, C=US
Signature Algorithm: rsassa_pkcs1v15
Valid From: 2026-10-08 16:55:08+00:00
Valid To: 2056-09-30 16:55:08+00:00
Issuer: CN=Android Debug, O=Android, C=US
Serial Number: 0x1
Hash Algorithm: sha1
md5: 79822b3157e224e1d4164f5de63bf87f
sha1: c7ef8a2cb9642cf7c701ee08c7a13e4f123324d6
sha256: 52ce760490d45ad0c08e8a7304c19f3c6706b9ae49e16f98a32b35146f80b2d0
sha512: 8c29cd75971db602ac51467b752a75cb50b4d75abb4de9523aee14b6076e1bfaa9874b2aaf6395d43abd30b500cd526f6c381e1a30cb3d7f10319d1adb044713
PublicKey Algorithm: rsa
Bit Size: 2048
Fingerprint: f4aa8438b350b9a383481194ac9a064c0f628938a49fa786170bd73fe4c32a3f
Found 1 unique certificates

APPLICATION PERMISSIONS

PERMISSION	STATUS	INFO	DESCRIPTION
android.permission.INTERNET	normal	full Internet access	Allows an application to create network sockets.
android.permission.CAMERA	dangerous	take pictures and videos	Allows application to take pictures and videos with the camera. This allows the application to collect images that the camera is seeing at any time.
android.permission.WRITE_EXTERNAL_STORAGE	dangerous	read/modify/delete external storage contents	Allows an application to write to external storage.
android.permission.ACCESS_FINE_LOCATION	dangerous	fine (GPS) location	Access fine location sources, such as the Global Positioning System on the phone, where available. Malicious applications can use this to determine where you are and may consume additional battery power.
android.permission.ACCESS_COARSE_LOCATION	dangerous	coarse (network-based) location	Access coarse location sources, such as the mobile network database, to determine an approximate phone location, where available. Malicious applications can use this to determine approximately where you are.
android.permission.ACCESS_NETWORK_STATE	normal	view network status	Allows an application to view the status of all networks.

APKID ANALYSIS

FILE	DETAILS

FILE	DETAILS
classes.dex	<table><tr><th>FINDINGS</th><th>DETAILS</th></tr><tr><td>Anti-VM Code</td><td>Build.MANUFACTURER check</td></tr><tr><td>Compiler</td><td>r8</td></tr></table>
classes2.dex	<table><tr><th>FINDINGS</th><th>DETAILS</th></tr><tr><td>Compiler</td><td>r8</td></tr></table>

NETWORK SECURITY

NO	SCOPE	SEVERITY	DESCRIPTION

CERTIFICATE ANALYSIS

HIGH: 1 | WARNING: 2 | INFO: 1

TITLE	SEVERITY	DESCRIPTION
Signed Application	info	Application is signed with a code signing certificate
Application vulnerable to Janus Vulnerability	warning	Application is signed with v1 signature scheme, making it vulnerable to Janus vulnerability on Android 5.0-8.0, if signed only with v1 signature scheme. Applications running on Android 5.0-7.0 signed with v1, and v2/v3 scheme is also vulnerable.
Application signed with debug certificate	high	Application signed with a debug certificate. Production application must not be shipped with a debug certificate.
Certificate algorithm might be vulnerable to hash collision	warning	Application is signed with SHA1withRSA. SHA1 hash algorithm is known to have collision issues. The manifest file indicates SHA256withRSA is in use.

MANIFEST ANALYSIS

HIGH: **3** | WARNING: **2** | INFO: **0** | SUPPRESSED: **0**

NO	ISSUE	SEVERITY	DESCRIPTION
1	App can be installed on a vulnerable unpatched Android version 6.0-6.0.1, [minSdk=23]	high	This application can be installed on an older version of android that has multiple unfixed vulnerabilities. These devices won't receive reasonable security updates from Google. Support an Android version => 10, API 29 to receive reasonable security updates.
2	Clear text traffic is Enabled For App[android:usesCleartextTraffic=true]	high	The app intends to use cleartext network traffic, such as cleartext HTTP, FTP stacks, DownloadManager, and MediaPlayer. The default value for apps that target API level 27 or lower is "true". Apps that target API level 28 or higher default to "false". The key reason for avoiding cleartext traffic is the lack of confidentiality, authenticity, and protections against tampering; a network attacker can eavesdrop on transmitted data and also modify it without being detected.

NO	ISSUE	SEVERITY	DESCRIPTION
3	Debug Enabled For App [android:debuggable=true]	high	Debugging was enabled on the app which makes it easier for reverse engineers to hook a debugger to it. This allows dumping a stack trace and accessing debugging helper classes.
4	Application Data can be Backed up [android:allowBackup=true]	warning	This flag allows anyone to backup your application data via adb. It allows users who have enabled USB debugging to copy application data off of the device.
5	Activity (com.example.watersos.Activities.DatosReporteActivity) is not Protected. [android:exported=true]	warning	An Activity is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device.

</> CODE ANALYSIS

HIGH: 1 | WARNING: 2 | INFO: 0 | SECURE: 0 | SUPPRESSED: 0

NO	ISSUE	SEVERITY	STANDARDS	FILES
1	IP Address disclosure	warning	CWE: CWE-200: Information Exposure OWASP MASVS: MSTG-CODE-2	com/example/watersos/Activities/CrearCuenta.java com/example/watersos/Activities/MainActivity.java com/example/watersos/JavaClass/EnviarDatos.java com/example/watersos/fragments/nav_Reportar.java com/example/watersos/fragments/nav_misReportes.java

NO	ISSUE	SEVERITY	STANDARDS	FILES
2	App uses SQLite Database and execute raw SQL query. Untrusted user input in raw SQL queries can cause SQL Injection. Also sensitive information should be encrypted and written to the database.	warning	CWE: CWE-89: Improper Neutralization of Special Elements used in an SQL Command ('SQL Injection') OWASP Top 10: M7: Client Code Quality	com/example/watersos/SQLite/AdminSQLiteOpenHelper.java
3	Debug configuration enabled. Production builds must not be debuggable.	high	CWE: CWE-919: Weaknesses in Mobile Applications OWASP Top 10: M1: Improper Platform Usage OWASP MASVS: MSTG-RESILIENCE-2	com/example/watersos/BuildConfig.java

NIAP ANALYSIS v1.3

NO	IDENTIFIER	REQUIREMENT	FEATURE	DESCRIPTION

BEHAVIOUR ANALYSIS

RULE ID	BEHAVIOUR	LABEL	FILES
00001	Initialize bitmap object and compress data (e.g. JPEG) into bitmap object	camera	com/example/watersos/SQLite/AdminSQLiteOpenHelper.java
00091	Retrieve data from broadcast	collection	com/example/watersos/fragments/nav_Reportar.java

ABUSED PERMISSIONS

TYPE	MATCHES	PERMISSIONS
Malware Permissions	6/25	android.permission.INTERNET, android.permission.CAMERA, android.permission.WRITE_EXTERNAL_STORAGE, android.permission.ACCESS_FINE_LOCATION, android.permission.ACCESS_COARSE_LOCATION, android.permission.ACCESS_NETWORK_STATE
Other Common Permissions	0/44	

Malware Permissions:
Top permissions that are widely abused by known malware.
Other Common Permissions:
Permissions that are commonly abused by known malware.

! OFAC SANCTIONED COUNTRIES

This app may communicate with the following OFAC sanctioned list of countries.

DOMAIN	COUNTRY/REGION

DOMAIN MALWARE CHECK

DOMAIN	STATUS	GEOLOCATION
192.168.1.66	ok	No Geolocation information available.

DOMAIN	STATUS	GEOLOCATION
weather.com	ok	**IP:** 184.86.165.205 **Country:** United States of America **Region:** Texas **City:** Dallas **Latitude:** 32.783058 **Longitude:** -96.806671 **View:** Google Map

EMAILS

EMAIL	FILE
android.studio@android.com	Android String Resource

HARDCODED SECRETS

POSSIBLE SECRETS
3e6720c1da23efaa42e23664474bc9860e41d18d0991a48e2b3fedf2a0f76f77

SCAN LOGS

Timestamp	Event	Error

2026-10-21 16:38:24	Generating Hashes	OK
2026-10-21 16:38:24	Extracting APK	OK
2026-10-21 16:38:24	Unzipping	OK
2026-10-21 16:38:24	Parsing APK with androguard	OK
2026-10-21 16:38:25	Extracting APK features using aapt/aapt2	OK
2026-10-21 16:38:25	Getting Hardcoded Certificates/Keystores	OK
2026-10-21 16:38:28	Parsing AndroidManifest.xml	OK
2026-10-21 16:38:28	Extracting Manifest Data	OK
2026-10-21 16:38:28	Manifest Analysis Started	OK
2026-10-21 16:38:28	Performing Static Analysis on: WaterSOS (com.example.watersos)	OK
2026-10-21 16:38:30	Fetching Details from Play Store: com.example.watersos	OK

2026-10-21 16:38:31	Checking for Malware Permissions	OK
2026-10-21 16:38:31	Fetching icon path	OK
2026-10-21 16:38:31	Library Binary Analysis Started	OK
2026-10-21 16:38:31	Reading Code Signing Certificate	OK
2026-10-21 16:38:32	Running APKiD 3.1.0	OK
2026-10-21 16:38:35	Detecting Trackers	OK
2026-10-21 16:38:36	Decompiling APK to Java with JADX	OK
2026-10-21 16:38:50	Converting DEX to Smali	OK
2026-10-21 16:38:50	Code Analysis Started on - java_source	OK
2026-10-21 16:38:51	Android SBOM Analysis Completed	OK
2026-10-21 16:38:52	Android SAST Completed	OK

2026-10-21 16:38:52	Android API Analysis Started	OK
2026-10-21 16:38:55	Android API Analysis Completed	OK
2026-10-21 16:38:56	Android Permission Mapping Started	OK
2026-10-21 16:38:58	Android Permission Mapping Completed	OK
2026-10-21 16:38:58	Android Behaviour Analysis Started	OK
2026-10-21 16:38:59	Android Behaviour Analysis Completed	OK
2026-10-21 16:38:59	Extracting Emails and URLs from Source Code	OK
2026-10-21 16:38:59	Email and URL Extraction Completed	OK
2026-10-21 16:38:59	Extracting String data from APK	OK
2026-10-21 16:38:59	Extracting String data from Code	OK
2026-10-21 16:38:59	Extracting String values and entropies from Code	OK

2026-10-21 16:39:00	Performing Malware check on extracted domains	OK
2026-10-21 16:39:06	Saving to Database	OK

Report Generated by - MobSF v4.5.4

Mobile Security Framework (MobSF) is an automated, all-in-one mobile application (Android/iOS/Windows) pen-testing, malware analysis and security assessment framework capable of performing static and dynamic analysis.

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